Keyu He

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Education

Carnegie Mellon University, Pittsburgh, PA — **May 2027**
M.S. in Intelligent Information Systems, School of Computer Science — **GPA: 4.14/4.33**

University of Southern California, Los Angeles, CA — **May 2025**
B.S. in Computer Science; B.A. in Applied & Computational Mathematics — **GPA: 3.98/4.00**
Minor in Artificial Intelligence Applications

Skills

Programming: Python, C++, TypeScript/JavaScript, Java, SQL, C, x86-64 Assembly
ML/Research: PyTorch, NumPy/Pandas, LLM/VLM evaluation, multi-agent systems, reinforcement learning
Systems/Tools: Databricks, AWS, RAG, LangChain, FAISS, Git, VS Code extensions, REST APIs, LaTeX

Experience

Adobe, San Jose, CA — May 2026 – Aug. 2026
Machine Learning Engineer Intern

- Shipped 5+ agent skills via an **agent skill generator** merged into the team's shared skill repo, turning rough developer requests into standards-compliant skills grounded in internal prior art and self-audited against the agentskills.io spec.
- Developed the systematic quality measure for the team's agent recommendation pipeline, a **reference-free 6-dimension rubric** combining LLM-as-judge with deterministic checks, reproducible to 0.01–0.05 per-dimension variance across repeated runs and surfacing two systematic weaknesses for the team to act on.
- Took that measure to production across the org: ported it onto the internal GenAI evaluation service as a reusable scorer and staged it end-to-end for two enterprise customers, on a four-stage Databricks pipeline built to run it.

CMind, Remote — May 2025 – Aug. 2025
Data Engineer Intern

- Built a LangChain-based RAG prototype over an internal analytics DB (chunking, embeddings, FAISS indexing/retrieval, LLM answer synthesis) with metadata filters and notebook/API utilities.
- Designed CEO-speech trait metrics and an auto-evaluation pipeline with custom rubrics, achieving Fleiss' κ of 0.89–0.91 against human annotations.

Publications

1. **Keyu He**, Xuhui Zhou, Maarten Sap. *SocialGym: Evaluating LLM Social Intelligence through Verifiable Multi-Agent Games*. Preprint, 2026. 10.48550/arXiv.2608.09128
   [rule-verifiable multi-agent environments; LLM orchestration; large-scale benchmarking; self-play design]
2. **Keyu He** et al. *Believing without Seeing: Quality Scores for Contextualizing Vision-Language Model Explanations*. ACL 2026. 10.48550/arXiv.2509.25844
   [VLM metric design; calibration; human-subject studies; statistical analysis]
3. **Keyu He***, Qianou Ma* et al. *RECAP: An End-to-End Platform for Capturing, Replaying, and Analyzing AI-Assisted Programming Interactions*. ACL Demo 2026. 10.48550/arXiv.2605.01104
   [VS Code instrumentation; full-stack TypeScript development; cloud data pipelines; human–AI interaction analysis]
4. Brihi Joshi*, **Keyu He*** et al. *ELI-Why: Evaluating the Pedagogical Utility of LLM Explanations*. Findings of ACL 2025. 10.48550/arXiv.2506.14200
   [benchmark design; human-subject studies; web-app implementation; multi-metric LLM evaluation]
5. Huihan Li*, Arnav Goel*, **Keyu He** et al. *Attributing Culture-Conditioned Generations to Pretraining Corpora*. ICLR 2025. 10.48550/arXiv.2412.20760.
   [large-scale corpus analysis; Python data engineering and visualization; survey server infrastructure design]

   * Equal contribution.